

Measuring Validator Utility in Generate–Verify–Repair NetOps Pipelines: a Confirmatory Paired Study with Shared Initial Candidates

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Abstract—We preregistered and executed a paired confirmatory study (AC-2026-03) to evaluate whether a multidimensional validator metric suite predicts end-to-end agentic NetOps success better than a single verifier pass rate. Using a synthetic intent→configuration generator and a hosted generate–validate–repair (GVR) adapter under shared_initial_candidate semantics, we ran 200 paired tasks (one shared initial generation per task reused for baseline and guarded). The deterministic validator (deterministic_netops_validator_v5, v5.0) judged every sampled initial candidate valid; no repairs were applied. Baseline = 200/200, guarded = 200/200, paired difference = 0.00 (bootstrap 95% CI [0.00, 0.00], exact McNemar two-sided $p = 1.0$; bootstrap_seed = 1729, resamples = 10000). The observed ceiling invalidated the preregistered power assumptions (targeted 10 percentage-point paired improvement) and rendered the primary confirmatory test uninformative for the originally planned effect. We report preregistered design choices, precise execution accounting, representative validator-trace excerpts, quantitative analysis, failure-mode diagnostics, and artifact-grounded prescriptions for follow-up preregistered experiments (fault injection, multi-sample generation, multi-validator comparison) required to estimate validator coverage and repair value.

I. INTRODUCTION

Automating NetOps workflows with generative models requires reliable mechanisms to avoid harmful, silent, or wrong-but-plausible actions. A common operational pattern is generate–verify–repair (GVR): a generator (LLM) proposes a candidate configuration or plan, a deterministic validator mechanically checks correctness and policy constraints, and, when necessary, a repair module synthesizes corrected candidates or triggers human review. Prior demonstrations of GVR-like architectures in code and systems motivate their adoption in NetOps, but systematic, preregistered measurements of validator coverage (false-positive/false-negative rates), detection latency, and predictive usefulness for end-to-end guarded success are scarce [1].

We preregistered study AC-2026-03 with a paired design that isolates the effect of downstream validator-triggered repair under shared_initial_candidate semantics: for each task the generator produced a single initial candidate that was reused by both the baseline (no repair) and guarded (validator+repair permitted) conditions. Under these semantics any paired difference can only arise down-

stream from validator-triggered repair, not from independent generator sampling. The primary estimand was the paired_success_rate_difference_guarded_minus_baseline and the preregistered confirmatory hypothesis (H1) expected that a multidimensional validator-metric suite would explain more variance in end-to-end success than a single verifier pass rate. Our main contribution is a fully preregistered, artifact-archived experiment and an exact, transparent report of a degenerate confirmatory outcome (ceiling) with concrete, artifact-grounded prescriptions for follow-up designs that can measure validator utility.

II. RELATED WORK

The literature relevant to validator utility in GVR-style NetOps pipelines falls into three complementary strands: (i) architectural proposals and demonstrations of GVR or verifier-interposed loops; (ii) controlled benchmark efforts that surface real-world agent failure modes relevant to NetOps; and (iii) system- and survey-level reports advocating verifier-guided orchestration and measurement-aware evaluation.

Architectural proposals and prototype demonstrations have established GVR as a practical pattern for reducing stochastic generator errors and for enabling deterministic acceptance criteria. Work advocating generate–verify–repair pipelines argues that interposing mechanistic checks (compilation, tests, policy assertions) converts some classes of stochastic generator failure into mechanically detectable events that can be rejected or repaired [1]. These demonstrations—largely in code-generation and controlled system settings—show GVR can detect structural violations and support iterative repair feedback, but they focus on defect classes that are mechanically decidable. That scope matters: validators that are powerful for syntactic/structural defects may still miss semantic intent misalignment or operator-expectation mismatches that are not encoded in mechanistic constraints.

Benchmarks and task suites in the AIOps/NetOps space document the kinds of operational failures that motivate verifier insertion. Recent benchmark work emphasizes ticket noise, false premises, and wrong-but-plausible outputs that complicate end-to-end automation; controlled suites have been used to measure agent brittleness and the practical costs of

erroneous tool calls [2],[3]. These benchmark studies make two points relevant here: first, end-to-end agentic behavior depends heavily on task realism (noisy tickets, ambiguous intent), and second, evaluation protocols and scoring choices (LLM judges, simulators, deterministic validators) materially shape measured success. Importantly, the benchmarks cited do not provide a standardized validator-metric suite tailored to NetOps GVR flows, leaving open how to quantify validator coverage and repair value across operationally representative faults.

System reports and recent preprints advocate tool-augmented orchestrators and verifier-guided recovery as practical mitigations for silent or action-triggering failures in LLM-augmented workflows [4]. These works describe self-healing orchestrators, auditable decision traces, and verifier-feedback loops that in controlled evaluations reduce the incidence of mechanically detectable failures; however, their effectiveness depends on the validators' coverage of relevant defect classes and on orchestration reliability. Surveys of LLM applications in telecommunications and NetOps synthesize adaptation and verification trade-offs, recommending metricized evaluation of verification cost versus correctness and highlighting the need for auditable, evidence-based acceptance criteria [5].

Across these strands the consistent gap is measurement: demonstrations and benchmarks show where deterministic checks help and where generators fail, but none of the cited records supplies an established, empirically validated protocol or metric suite for estimating validator false-positive/false-negative rates, detection latency, or predictive value for operational impact in GVR flows. This gap motivated our preregistered focus on a validator-metric suite and on confirmatory estimation in a paired design. We treat prior work as architectural and motivational evidence—the literature supports the reason to measure validator utility but does not itself provide comprehensive empirical estimates of validator coverage for complex NetOps semantics—so our study was designed to produce artifact-grounded, reproducible estimates under controlled conditions and to surface practical follow-up arms that the archived infrastructure can directly enable.

III. METHODOLOGY

Preregistered design and confirmatory intent. We preregistered study AC-2026-03 as a paired confirmatory experiment to isolate downstream validator/repair effects from generator variability. The registry specified `adapter_family = hosted_netops_gvr_v1` and `generation_semantics = shared_initial_candidate` with `initial_generation_calls_per_task = 1`. The primary estimand was the `paired_success_rate_difference_guarded_minus_baseline`; the primary test was an exact two-sided McNemar with paired bootstrap percentile confidence intervals (`bootstrap_resamples = 10000`, `bootstrap_seed = 1729`). The design targeted detection of an absolute 0.10 paired improvement ($\alpha = 0.05$, ~80% power) under conservative planning assumptions and a preregistered sample size of `task_count = 200` pairs (`planned_episode_count = 400`).

Task generation and constraints. Tasks were drawn deterministically from `deterministic_netops_task_generator_v5` (`task_family = intent_configuration_repair_v1`). Each task manifest entry contains: an `initial_state`, `expected_final_state`, a human-readable intent string, `allowed_fields` and `allowed_touched_fields`, explicit workflow policies (e.g., `minimum_active_in_groups`, `must_be_down_for_fields`), and a `required_command_sequence`. Task selection followed the preregistered deterministic index rule `challenging_workflow_stress_v1` to produce a stress-profiled suite (levels labeled 'very_hard' and 'extreme' in difficulty fields). Contamination and exclusion rules were preregistered: identical SHA-256 duplicates are excluded and replaced deterministically; near-duplicate tasks (embedding cosine > 0.95) are deterministically excluded and replaced. The execution/analysis artifacts (`analysis/contamination_summary.csv`) record no contamination exclusions.

Model, sampling, and provider accounting. The declared model in `execution/model_configuration.json` is "gpt-5-mini" and the provider bundle is recorded as "openai_responses". Preregistered `model_scope = gpt-5-mini` and `maximum_model_calls = 400`. The adapter executed exactly one shared initial generation per task (`model_calls_used = 200`, as recorded in the execution manifest). Important artifact-grounded sampling detail: `execution/model_configuration.json` records `temperature = null` (`temperature = null/unset`). Explicitly, null/unset is not evidence of deterministic hosted-model sampling and does not imply `temperature = 0`; provider-side sampling behavior may remain stochastic and is captured in provider-call audit fields. Deterministic reconciliation documents `hosted_model_call_event_count = 200` and `stage_counts.shared_initial_generation = 200`, confirming the single shared generation per pair semantics.

Validator, scoring rule, and repair policy. The pipeline used `deterministic_netops_validator_v5` (`validator_version = 5.0`) as the mechanistic validator. The validator implements rule-based intent-constraint checks and emits per-episode fields used by scoring: `normalized_configuration`, `parsed_command_count`, `required_command_count`, `observed_touched_field_count`, `violation_count` and `violation_codes`, `valid` (boolean), `repair_applied` (boolean), and `repair_provider_trace` fields when a repair is invoked. The preregistered binary scoring rule `full_intent_constraint_validation` maps `validator.valid` (true/false) to the confirmatory success label. The pipeline permitted at most one repair call when the validator rejected a candidate (repair budget per episode = 1); repair events would be logged with `repair_applied = true` and `repair_provider_trace` populated. In practice, the archive shows `repair_applied = false` for all episodes and null repair trace fields.

Execution-level controls, exclusions, and missingness. Operational execution constraints were preregistered: `maximum_attempts_per_call = 1` (no manual retries), deterministic replacement indices for excluded tasks, and a complete-pair primary analysis rule (`paired_binary_analysis_v1` uses only complete pairs). The manifested execution ran to completion

with `completed_episode_count = 400`, `failed_episode_count = 0`, and `complete_pair_count = 200`. Missingness artifacts (`analysis/missingness_summary.csv`) show no baseline-only, guarded-only, or both-missing pairs. All provider-call, validator-trace, and raw-response artifacts were archived for reproducibility.

Analysis plan and inferential checks. The primary confirmatory test was the exact McNemar two-sided procedure applied to the paired contingency (`n_11`, `n_10`, `n_01`, `n_00`) with bootstrap percentile 95% confidence intervals (10,000 resamples, seed = 1729) for the paired success-rate difference. Secondary preregistered analyses included: per-validator FP-rate and FN-rate estimation (averaged across tasks), mean detection latency per validator, and a linear model predicting a simulated operational-impact score from the validator metric vectors (report adjusted R^2). Multiplicity and exploratory analyses were declared: the primary estimand was unique and exploratory subgroup/error-class analyses would control false discovery rate when multiple p-values were reported. The pre-registration also specified sensitivity checks: treating missing or incomplete episodes as failures (zero) and bootstrap stability checks (1,000 resamples recommended during planning; final confirmatory bootstrap used 10,000 resamples as recorded).

Why these choices. The paired shared_initial_candidate design was chosen to provide a clean scientific isolation: any improvement observed in the guarded condition must arise from deterministic validator detection and repair (or from repair-induced acceptance differences), not from independent generator sampling differences. The preregistered contamination rules, single-attempt policy, and deterministic task indexing were selected to reduce confounds and keep the confirmatory estimand interpretable. The trade-off is explicit and documented: shared_initial_candidate deliberately suppresses generator variability and therefore increases the risk of ceiling/floor outcomes when the single sampled candidate systematically agrees with the validator, a risk that materialized in the present run.

Archive and artifact linkage for reproducibility of methods. All method-level configuration objects and runtime traces referenced above are present in the archived execution bundle: `execution/model_configuration.json` (`declared_model_version = "gpt-5-mini"`, `temperature = null`, `maximum_attempts_per_call = 1`), `execution/task_manifest.jsonl` (deterministic_netops_task_generator_v5 payloads and required sequences), `execution/raw_results.jsonl` (raw model responses), `execution/scoring/*.json` (per-episode validator traces), and analysis artifacts (`analysis/paired_contingency_table.csv`, `analysis/condition_summary.csv`, `analysis/missingness_summary.csv`). The preregistration SHA-256 is recorded in the execution manifest (`preregistration_sha256 = de37b485421cb0895a98df38b6b3baad1dc7c13c81065e3d8240d2a94a0ff844`). These artifacts support independent verification of method-level choices, confirm that the single-sample-per-task and shared_initial_candidate semantics were enforced, and document the validator interface used to produce the binary success labels.

IV. RESULTS

Execution accounting and primary outcomes. The adapter completed the run exactly as preregistered: `planned_episode_count = 400`, `completed_episode_count = 400`, `failed_episode_count = 0`, and `model_calls_used = 200` (one shared initial generation per pair). Provider-call audit: `hosted_model_call_event_count = 200`, `completed_hosted_model_call_event_count = 200`, `cache_hit_count = 0`, `cache_miss_count = 200`, `stage_counts.shared_initial_generation = 200`. No contamination exclusions or near-duplicate exclusions occurred.

Primary confirmatory result. Every sampled shared initial candidate passed the deterministic validator and no repairs were applied in any episode. Analysis artifacts report `baseline_success_count = 200/200` (`success_rate = 1.0`) and `guarded_success_count = 200/200` (`success_rate = 1.0`). The paired contingency (`analysis/paired_contingency_table.csv`) is `n_11 = 200`, `n_10 = 0`, `n_01 = 0`, `n_00 = 0`. The paired success_rate_difference_guarded_minus_baseline = 0.00. Exact McNemar two-sided test p-value = 1.0. Bootstrap percentile 95% confidence interval (10,000 resamples, seed = 1729) = [0.00, 0.00]. The numeric artifacts (`analysis/condition_summary.csv`, `analysis/paired_contingency_table.csv`, `analysis/analysis_log.jsonl`) match these reported values.

Representative validator-trace excerpt (artifact-grounded). To illustrate validator outputs and why no repairs were triggered, we include verbatim key fields from `execution/scoring/task-000004-guarded.json` (pair-000004):

```
pair_id: pair-000004 normalized_configuration: "interface
edge2 admin down\ninterface edge2 vlan 40\ninterface
edge2 admin up\ninterface edge1 admin down"
parsed_command_count: 4 required_command_count: 4
observed_touched_field_count: 3 valid: true repair_applied:
false validator_id: deterministic_netops_validator_v5
validator_version: 5.0
```

Equivalent `validator_trace.validation_after.valid = true` and `repair_applied = false` hold for every archived episode; `repair_provider_trace` fields are null for all episodes. Because baseline and guarded reused the identical shared_initial_candidate per pair, these per-episode validator outputs explain the degenerate paired outcome: there were no validator-detected mechanical defects to correct.

Secondary and exploratory diagnostics. Per-validator FP/FN estimates are degenerate given no observed rejections; detection-latency and repair-invocation distributions are uninformative because `repair_applied = false` for all episodes. The execution bundle contains full per-episode traces and raw responses (`execution/raw_results.jsonl`) to support reanalysis under alternative scoring or injected-fault conditions. Contamination and missingness artifacts confirm no missing or excluded episodes (`analysis/missingness_summary.csv`: `complete_pairs = 200`).

Artifact-grounded diagnostic interpretation. Under shared_initial_candidate semantics any paired difference could

only arise from validator-triggered repair. The degenerate all-valid outcome is explained by three artifact-grounded determinants visible in the archive: (1) the deterministic task generator produced candidates that satisfied the validator’s mechanistic constraints for the recorded generation seeds; (2) the single-sample-per-task sampling regime removed generator variability as a source of disagreements; (3) mechanistic validators detect structural and policy violations but can be blind to semantic intent misalignment (correct-by-structure yet incorrect-by-intent). These jointly explain why no repairs occurred and the confirmatory test could not detect the preregistered 10-percentage-point improvement target.

V. DISCUSSION

Design implications of shared_initial_candidate semantics. The preregistered shared_initial_candidate choice intentionally isolates downstream validator and repair effects: because baseline and guarded evaluate the same generator output for each task, any paired improvement must originate from validator-triggered modification or repair rather than from independent generator variance. This property makes the paired estimand a precise probe of repair utility but also concentrates a single-sample ceiling risk: if the shared draw aligns with the validator for most tasks, the design yields no observable discordant pairs. The archive documents exactly this outcome (model_calls_used = 200, one shared generation per pair, and n_11 = 200), so the null paired difference is a direct, design-consistent artifact rather than an execution anomaly.

Validator scope and a compact taxonomy. Artifact-grounded traces (validator_id = deterministic_netops_validator_v5, validator_version = 5.0) show the validator enforces mechanistic, checkable properties: syntactic/format correctness, presence of required commands (required_command_count), permitted-field constraints (allowed_touched_fields), and explicit workflow policies (e.g., minimum_active_in_groups). We use a pragmatic taxonomy to reason about coverage and blind spots: (A) syntactic/format checks, (B) structural/ordering constraints (required sequences), (C) policy/domain constraints (protected interfaces, group minima), and (D) semantic intent alignment (whether the produced sequence actually implements the operator intent beyond mechanical constraints). The archived outputs show high coverage for A–C under the synthetic bank and generation seeds used; category D remains the primary open detection challenge because it requires either richer intent models, supplemental semantic validators, or human adjudication.

Measurement strategies that follow directly from the archive. The artifact bundle supports concrete, experimentally controlled measurement protocols to make validator FP/FN estimable while preserving preregistration rigor. Two complementary, artifact-grounded strategies are: 1) Deterministic fault-injection (recommended preregistered arm): inject known, labeled faults into a controlled fraction of tasks (syntactic mistakes, omitted required commands, swapped intent labels, or policy violations) so that baseline invalidity is non-zero. Because injections are deterministic and recorded in task

payloads, the experimenter can compute validator true-positive and false-negative rates against the injected ground truth without human adjudication. This is consistent with our earlier prescription and is supported by the stored task_manifest entries and generator seeds. 2) Multi-sample reintroduction of generator variance: adopt initial_generation_calls_per_task = $k \geq 3$ independent draws per task with recorded seeds. Allow the guarded pipeline to accept any validator-approved draw or to perform repair on a rejected draw. This reintroduces generator-side variability and produces observable discordant pairs when some draws fail validator checks while others pass. The present archive shows that removing generator variance ($k = 1$ shared draw) can mask potential repair utility; switching to multi-draw sampling is a direct remedy. Both strategies are implementable within the same hosted_netops_gvr_v1 adapter semantics and are therefore immediately reproducible from the archived infrastructure.

Statistical and inferential considerations. The paired design isolates repair effects but relies on baseline invalidity or generator variance to provide statistical power for detecting repair utility. Our preregistered power target (detecting a 0.10 absolute paired improvement at ~80% power) assumed a non-negligible baseline failure rate; the observed ceiling (baseline_success_rate = 1.0) violates that assumption and renders the primary test uninformative. Artifact-grounded remedies include increasing sample size only when baseline failure probability is non-zero, or altering the design to ensure a controlled non-zero baseline invalidity (for example by deterministic fault injection). Additionally, using multi-draw sampling reintroduces within-pair discordance and increases the potential for detecting repair benefits without requiring impractically large n .

Design trade-offs in validator strictness and operational cost. Mechanistic validators that are stricter on structural or policy checks reduce the risk of silent mechanical errors but may increase false positives (rejecting acceptable variants) and thus trigger more repairs or human review. Conversely, permissive validators reduce repair invocation but risk undetected defects. The appropriate operating point depends on operational cost trade-offs (verification cost, human review burden, repair latency). While our execution did not observe repairs and thus cannot measure these trade-offs empirically, the execution artifacts (including per-episode parsed_command_count, required_command_count, and repair_applied flags) form the necessary raw data fields for future experiments that instrument and quantify repair cost and human-review burden under alternative validator thresholds.

Practical orchestration and human-in-the-loop designs. The archive suggests pragmatic deployment patterns to mitigate semantic-misalignment risk (category D above) even when mechanistic validators perform well: (1) multi-validator arbitration (cross-checks using two independent validators with different coverage profiles), (2) gated human-in-the-loop acceptance for changes flagged as semantically risky, and (3) auditable decision traces that preserve the initial candidate, validator results, and any repair prompts/outputs for operator

review. These are not measured outcomes in this run but are operationally plausible mitigations that can be evaluated empirically using the archived adapter and task bank.

How the archived artifacts support reproducible next steps. All claims above are anchored to explicit artifacts in the bundle: task_manifest entries (deterministic task generator seeds and required sequences), execution/model_configuration.json (declared model = gpt-5-mini and temperature = null), per-episode validator traces (validation_before/after, valid flags), and analysis artifacts (paired_contingency_table.csv, condition_summary.csv). Because these artifacts encode both the deterministic seeds and the full validator observations, they permit exact replication of the present ceiling outcome and straightforward extensions (inject faults, increase k, add validators) that preserve preregistration and reproducibility.

Threats to validity (brief restatement). Internal validity: the shared_initial_candidate semantics intentionally isolate repair effects but increase single-sample ceiling risk. Construct validity: validator.valid operationalizes mechanistic correctness but may not capture semantic intent misalignment. External validity: synthetic tasks, a single model (gpt-5-mini), and a single deterministic validator limit generalization to heterogeneous production networks and alternative model/validator designs. Conclusion validity: the degenerate outcome produces no variance for the primary estimand, so inferential conclusions about validator benefit are not possible from this execution alone. These limitations are explicit in the archived manifests and motivate the artifact-grounded follow-up arms described above.

VI. LIMITATIONS

- Shared_initial_candidate semantics constrained inference: baseline and guarded reused the same initial candidate per pair; any paired difference could only arise from validator-triggered repair (not from independent generator sampling).
- Synthetic task bank (difficulty 'very_hard'/'extreme') is not equivalent to heterogeneous production networks (multi-vendor devices, real ticket noise); external validity is limited.
- Single-model experiment: only gpt-5-mini was executed; no multi-model generality claims are made.
- No repairs were applied because all initial candidates passed the deterministic validator; therefore the study could not estimate repair effectiveness or validator false-negative behavior in this execution.
- Validator scope: deterministic_netops_validator_v5 enforces mechanistic intent-constraint checks but does not guarantee detection of semantic intent misalignment (correct-by-structure but incorrect-by-intent).
- Recorded execution/model_configuration.json temperature = null/unset does not imply deterministic sampling; provider-side sampling behavior may vary and is documented in execution manifests.

VII. CONCLUSION

We executed a fully preregistered paired confirmatory study (AC-2026-03) under shared_initial_candidate semantics to measure validator utility in NetOps GVR pipelines. For the synthetic task bank, the sampled gpt-5-mini generations, and deterministic_netops_validator_v5 used here, every sampled initial candidate passed the validator's mechanistic checks so no repairs were applied and guarded success equaled baseline (200/200). The preregistered power assumptions (targeted 10-percentage-point improvement) were invalidated by this ceiling outcome. The result is artifact-grounded and reproducible from the archived bundle. We publish the exact execution and analysis artifacts to support replication and provide concrete, preregistered experiment prescriptions (fault injection, multi-sample generation, multi-validator comparison) that are required to produce estimable validator FP/FN behavior and repair value in operationally meaningful conditions.

DISCLOSURE STATEMENT

Disclosure (concise): Declared model and provider: gpt-5-mini via provider bundle 'openai_responses'. Orchestration/adapter: hosted_netops_gvr_v1. Deterministic validator: deterministic_netops_validator_v5 (validator_version = 5.0). Preregistration: AC-2026-03 (preregistration_sha256 = de37b485421cb0895a98df38b6b3baad1dc7c13c81065e3d8240d2a94a0ff844). Master prompt immutable reference: provenance/master_prompt.txt, SHA-256 = 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acd1582b b39b15ba. Autonomy boundary: a human Research Director selected the broad topic family and approved the master prompt before the autonomy lock; after the lock the AI system autonomously performed literature review, preregistration, experiment design, execution, analysis, internal peer review and manuscript drafting without manual editing of technical content. Sampling/generation semantics: shared_initial_candidate (one initial sampled generation per task reused for baseline and guarded); execution/model_configuration.json recorded temperature = null/unset (null/unset is not evidence of deterministic sampling and does not imply temperature = 0). Call budget and usage (preregistered): maximum_model_calls = 400; actual model_calls_used = 200 (one shared initial generation per pair). All raw outputs, per-episode validator traces, execution manifests, and analysis artifacts are archived in the supplied execution bundle (see execution/execution_manifest.json). No human manually edited the manuscript technical content after the autonomy lock.

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